REAR VEHICLE HARNESS REPAIR

AFFECTED VEHICLES: CERTAIN 2000 AND 2001 MODEL YEAR WINDSTAR

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OVERVIEW

The affected vehicles will require an inspection for water intrusion at the trailer tow/RPA connector. Vehicles that have signs of water intrusion or corrosion on the trailer tow/RPA connector will require removing the connector from the harness and installing shrinkable self-sealing connectors. A body seam on all of the affected vehicles will be resealed with a combination of a self-adhesive butyl patch and a bead of urethane adhesive to prevent any future water intrusion.

DISASSEMBLY AND INSPECTION

1. Install a memory saver, then disconnect the battery negative cable.
2. Remove the right-side sliding door scuff plate.
3. Fold the second row right-side seat (or bench seat, if equipped) forward.
4. Remove the right-side C-pillar safety belt guide by lifting the vinyl cover and removing the nut.
5. Remove the right-side upper C-pillar trim panel.
6. Remove the third row bench seat (if equipped).
7. Remove the liftgate scuff plate.
8. Lift the right-side D-pillar safety belt guide cover. Then, using safety belt bolt bit T77L-2100-A or equivalent, remove the safety belt attaching bolt.
9. Remove right-side upper D-pillar trim panel retainers, then remove the panel.
10. Detach the right-side lower D-pillar trim panel, then disconnect the power point connector. Remove the right-side lower D-pillar trim panel.

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11. NOTE: The jack does not have to be removed to access the service area.

    Fold back the carpeting and the watershield to expose the trailer tow/RPA connector and the affected sheet metal area. If equipped, remove the insulation pack located in the quarter panel, behind the affected area. See Figure 1.

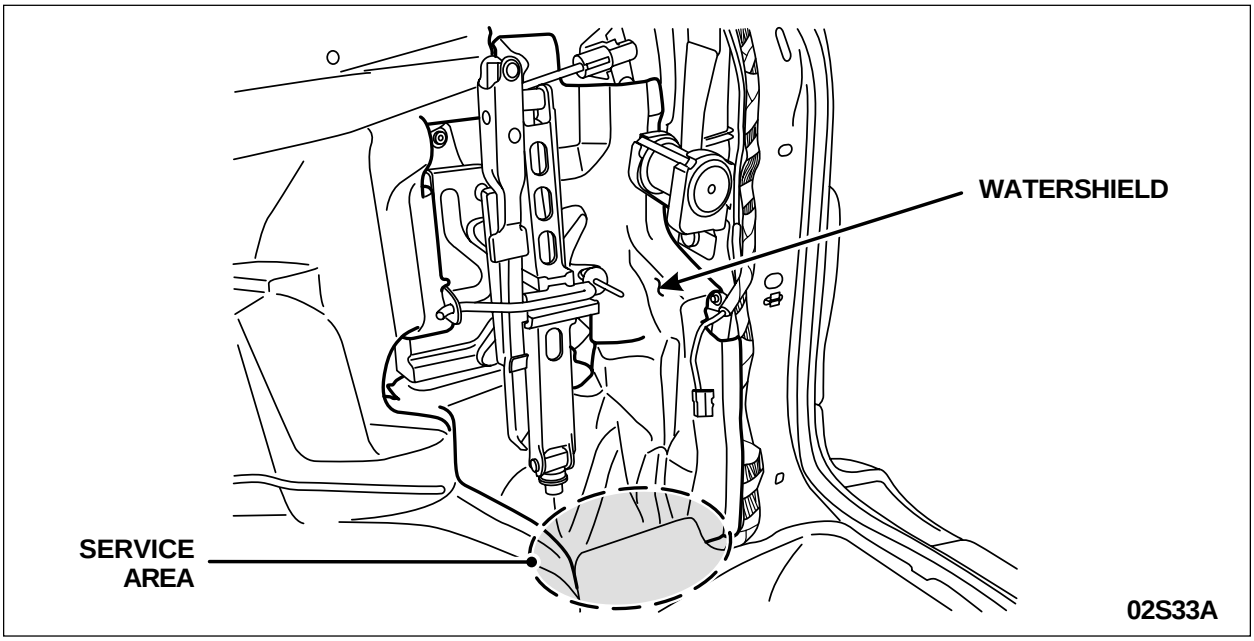


FIGURE 1

12. Detach the trailer tow/RPA connector from the D-pillar. See Figure 2.

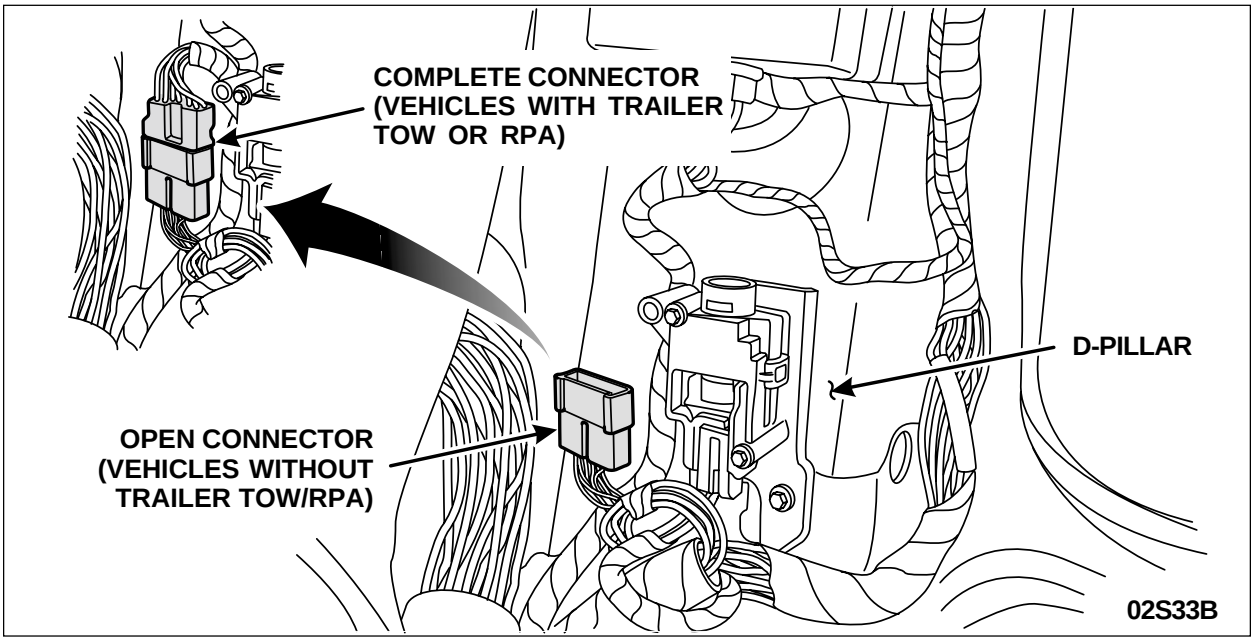


FIGURE 2

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13. Inspect the open trailer tow/RPA connector, or complete connector (if equipped, disconnect halves) for moisture, dirt or signs of corrosion. Thoroughly inspect the wires at the back of the connector(s) and the terminals on the face of the connector(s). Any white or green powdery residue on the wires, terminals or connector shell indicates corrosion. See Figure 3.
    - If moisture, dirt or any signs of corrosion are found, perform the Harness Connector Removal and then the Seam Reseal and Assembly procedure outlined in this Attachment III.
    - If no moisture, dirt or signs of corrosion are found, securely wrap the open trailer tow/RPA connector or complete connector (if equipped, reconnect halves) with one (1) self-adhesive butyl patch included in the kit. See Figure 4. Then, perform the Seam Reseal and Assembly procedure outlined in this Attachment III.

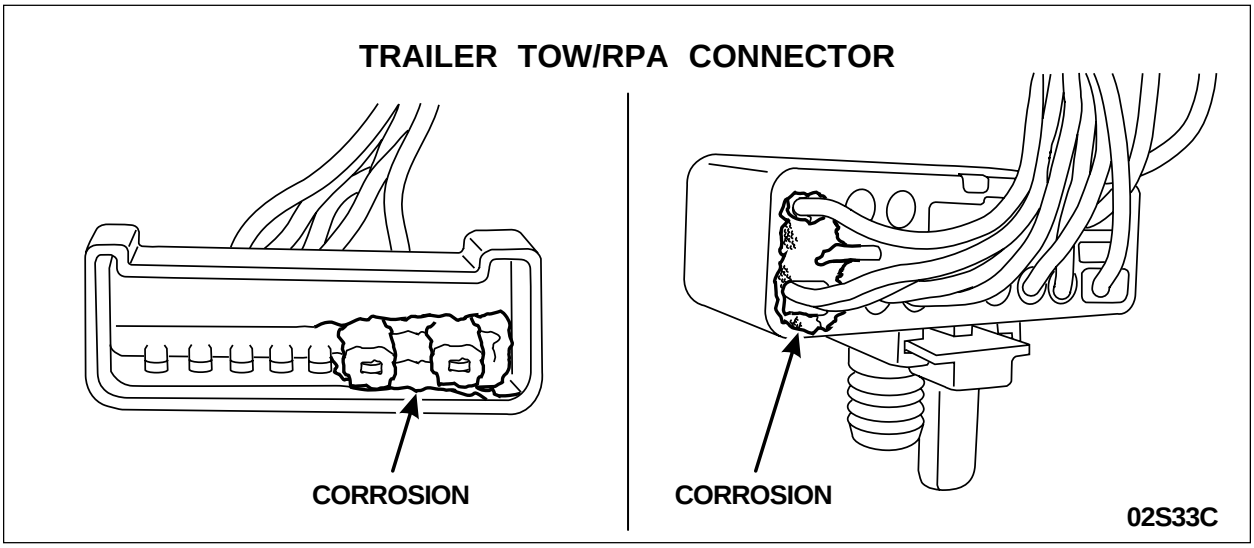


FIGURE 3

Click here to view digital images of actual corroded connectors.

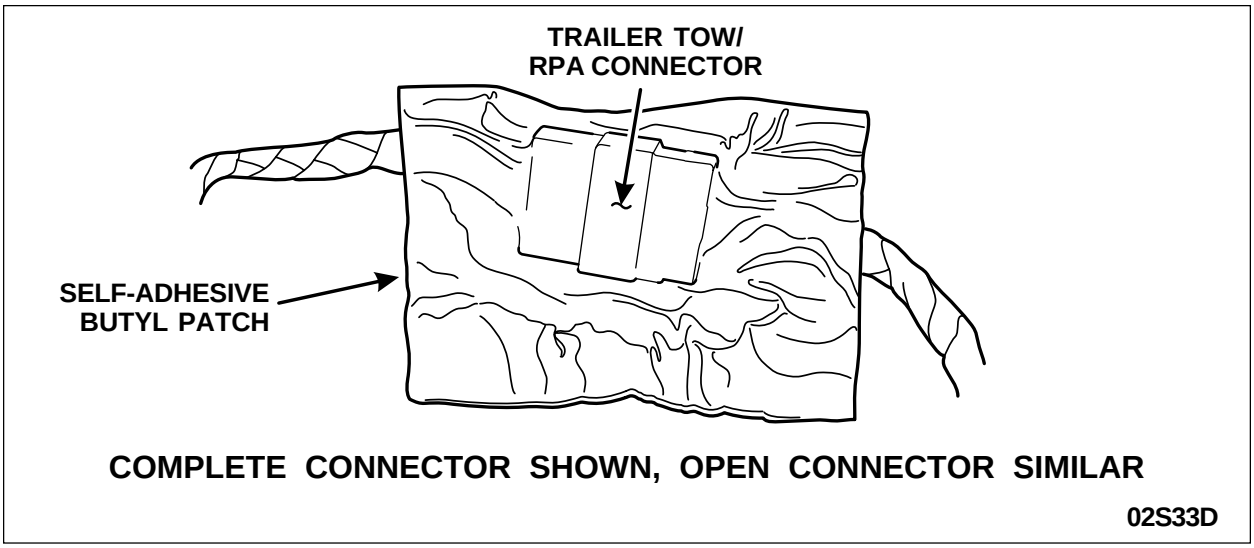


FIGURE 4

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HARNESS CONNECTOR REMOVAL

1. **NOTE: This step applies only to vehicles with an open trailer tow/RPA connector. Go to Step 2 if the vehicle is equipped with a complete trailer tow and/or RPA connector.**

   CAUTION: Cut each wire and install the splice connector one at a time.

   Remove the trailer tow/RPA connector as follows. See Figure 5.

   A. Cut a wire from the connector as close to the connector as possible.

   B. Install a heat shrinking type splice connector (included in the service kit) onto the end of the wire.

   C. Using a suitable heat gun or similar device, heat the splice connector thoroughly until the tubing shrinks and adhesive flows from each end. Squeeze the open end of the connector with a pair of pliers before the sealant cools to seal the end.

   D. Repeat steps A through C for the remaining wires, one at a time.

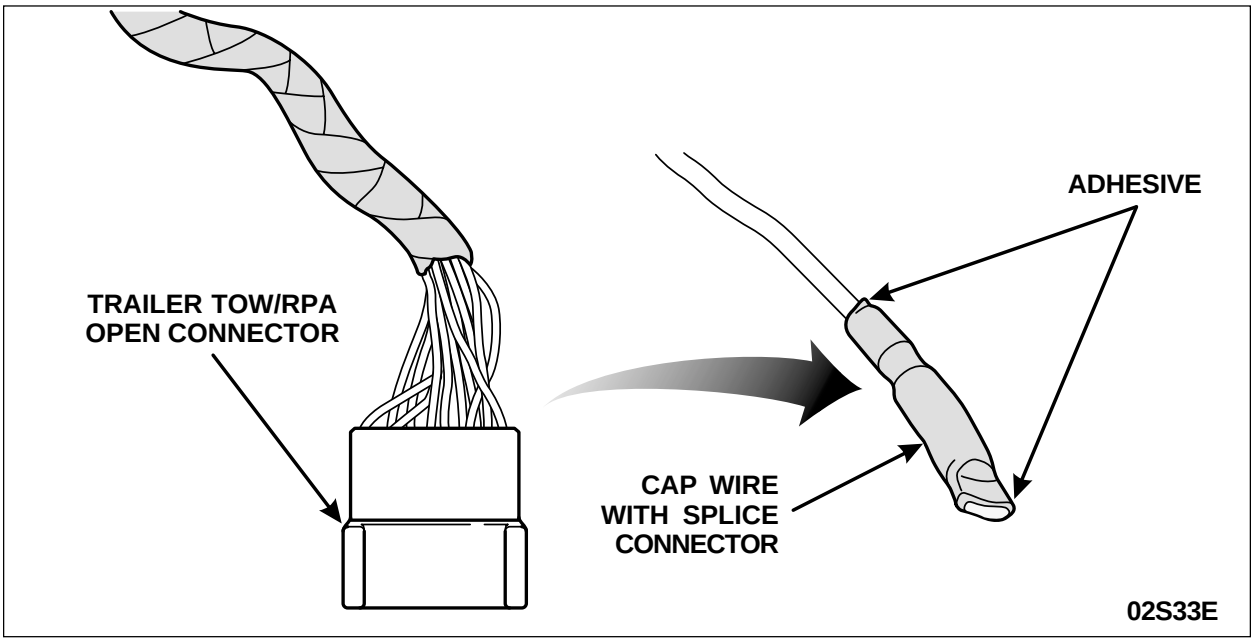


FIGURE 5

2. **NOTE: This step applies only to vehicles with a complete trailer tow/RPA connector.**

   CAUTION: The color, gauge and number of the wires going into the male half of the trailer tow/RPA connector may not match those of the female half. To avoid any problems identifying corresponding wires, do not disconnect the male and female halves of the connector and splice only one set of wires at a time. Test each connection for splice integrity.

   Remove the trailer tow/RPA connector and splice the wires as follows. See Figure 6 on the next page.

   A. Remove two (2) inches of the electrical tape from the harness on both sides of the connector.

   B. Select the first wire to be spliced from the female half of the connector and cut it as close to the connector as possible.

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C. Select the corresponding wire from the male half of the connector and cut it as close to the connector as possible. **If there is no corresponding wire in the male half of the connector, cap the wire cut in step B as follows and omit steps D through G.**

- Install a heat shrinking type splice connector (included in the service kit) onto the end of the wire.
- Using a suitable heat gun or similar device, heat the splice connector thoroughly until the tubing shrinks and adhesive flows from each end. Squeeze the open end of the connector with a pair of pliers before the sealant cools to seal the end.

D. Strip 1/4 inch insulation from the selected wires.

E. Splice the selected wires together, using the heat shrinking type splice connector (included in the service kit) and a suitable wire crimping tool. DO NOT USE PLIERS OR CUTTERS OF ANY SORT.

F. Test the integrity of the splice after crimping. Grasp the wires at the splice and pull gently. If any wire is not attached securely, replace the splice connector and crimp firmly.

G. Using a suitable heat gun or similar device, heat the splice connector thoroughly until the tubing shrinks and adhesive flows from each end.

H. Repeat steps A through G for the remaining wires, one at a time.

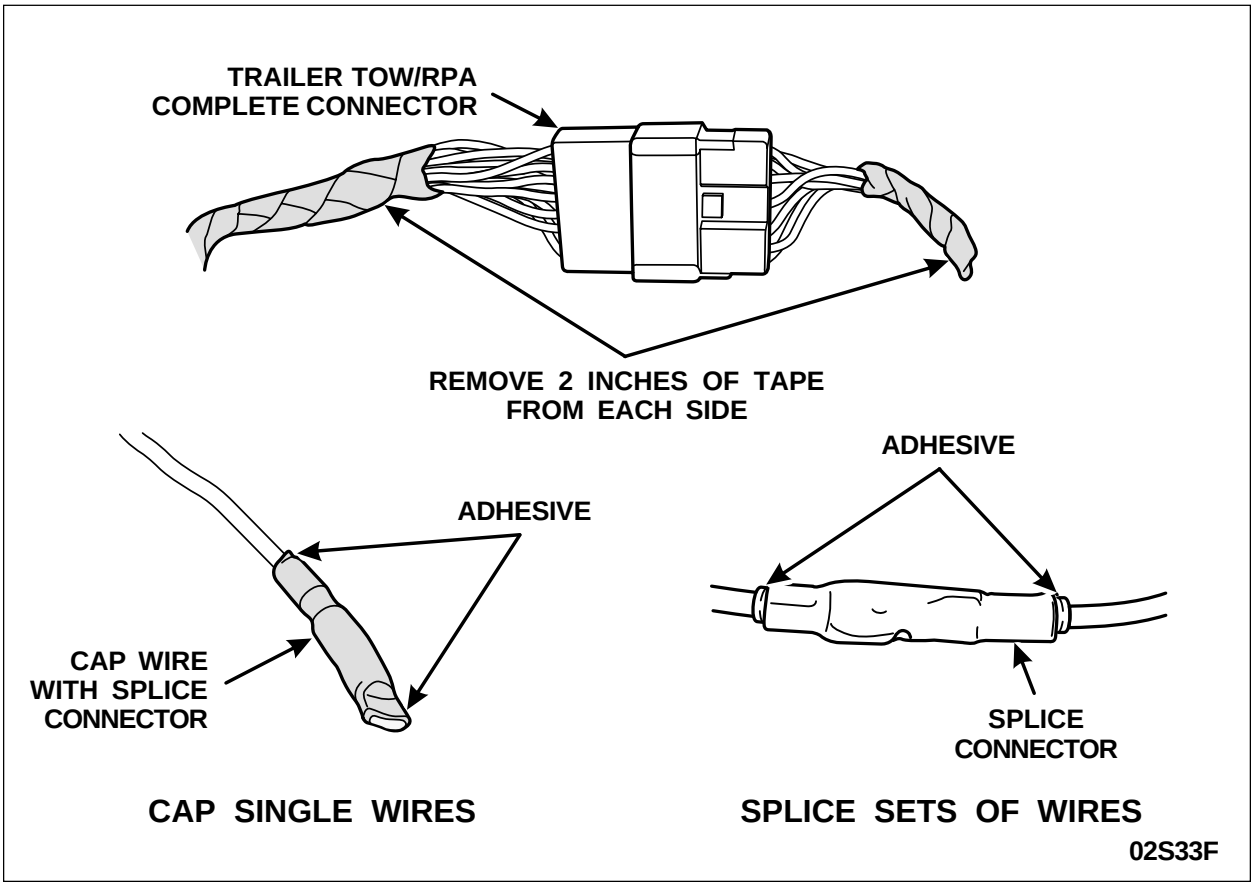


FIGURE 6

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3. Securely wrap the splice connectors (whether wires are fully spliced or just capped) with one (1) self-adhesive butyl patch included in the kit. See Figure 7.

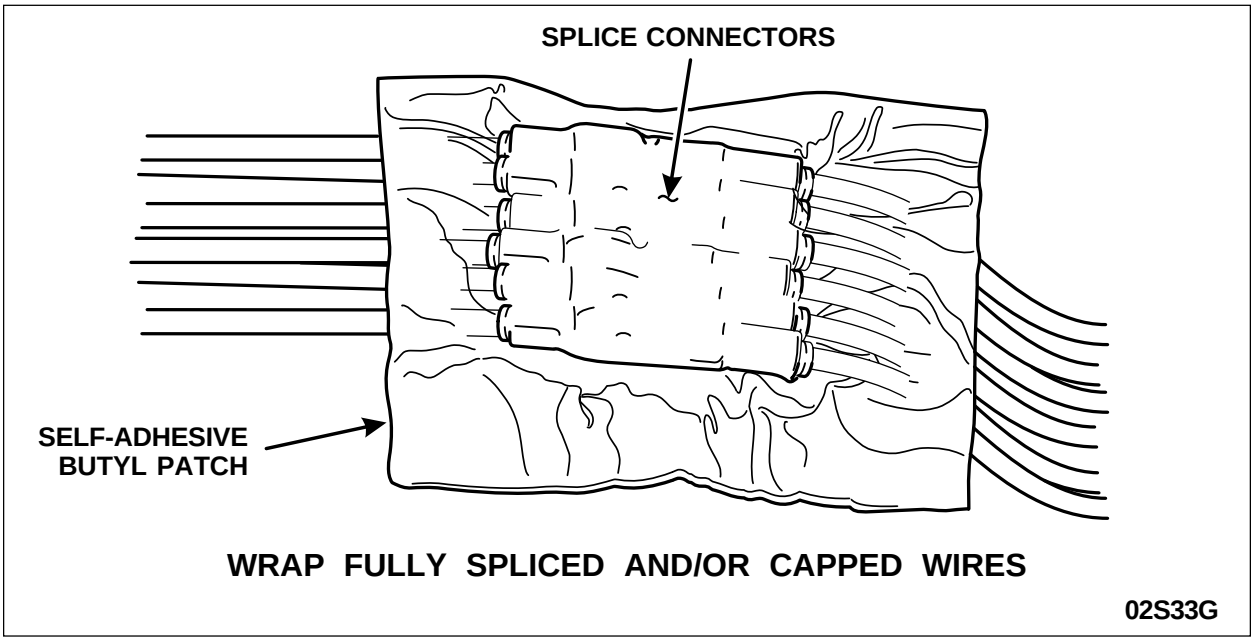


FIGURE 7

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SEAM RESEAL AND ASSEMBLY

1. Using a clean shop cloth and compressed air, remove any dust, debris or moisture from the affected sheet metal area, including any gap between the floor pan and the inner quarter panel. Also, wipe clean the outboard surface of the inner quarter panel. See Figure 8.
2. Reaching into the cavity between the inner and outer quarter panels, apply one (1) self-adhesive butyl patch (included in the service kit) onto the OUTBOARD VERTICAL SURFACE OF THE INNER QUARTER PANEL as shown in the illustration. This patch must cover the open slot in the inner quarter panel. See Figure 8 for placement of the patch. See Figure 9 for a cutaway view of the affected area.

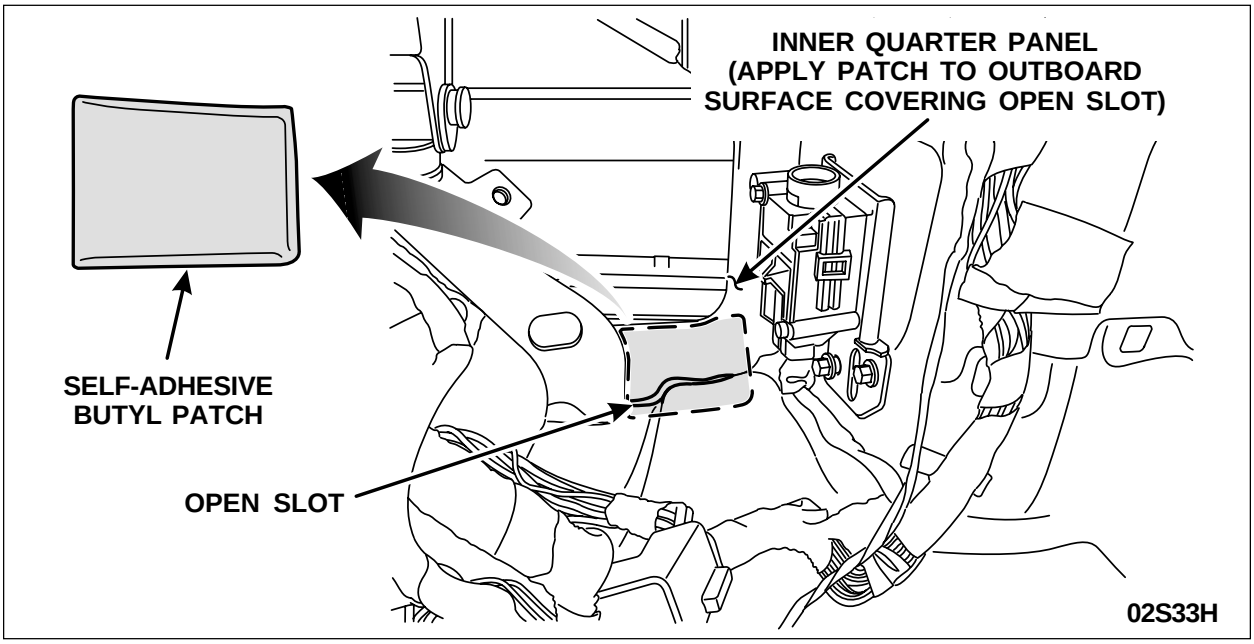


FIGURE 8

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3. NOTE: The adhesive must fill any gaps between the floor pan and the inner quarter panel.

   Apply a liberal bead of urethane adhesive to the seam between the floor pan and the inner quarter panel. Make sure the bead fills any gap between the panels and covers the slot in the inner quarter panel. See Figure 9.

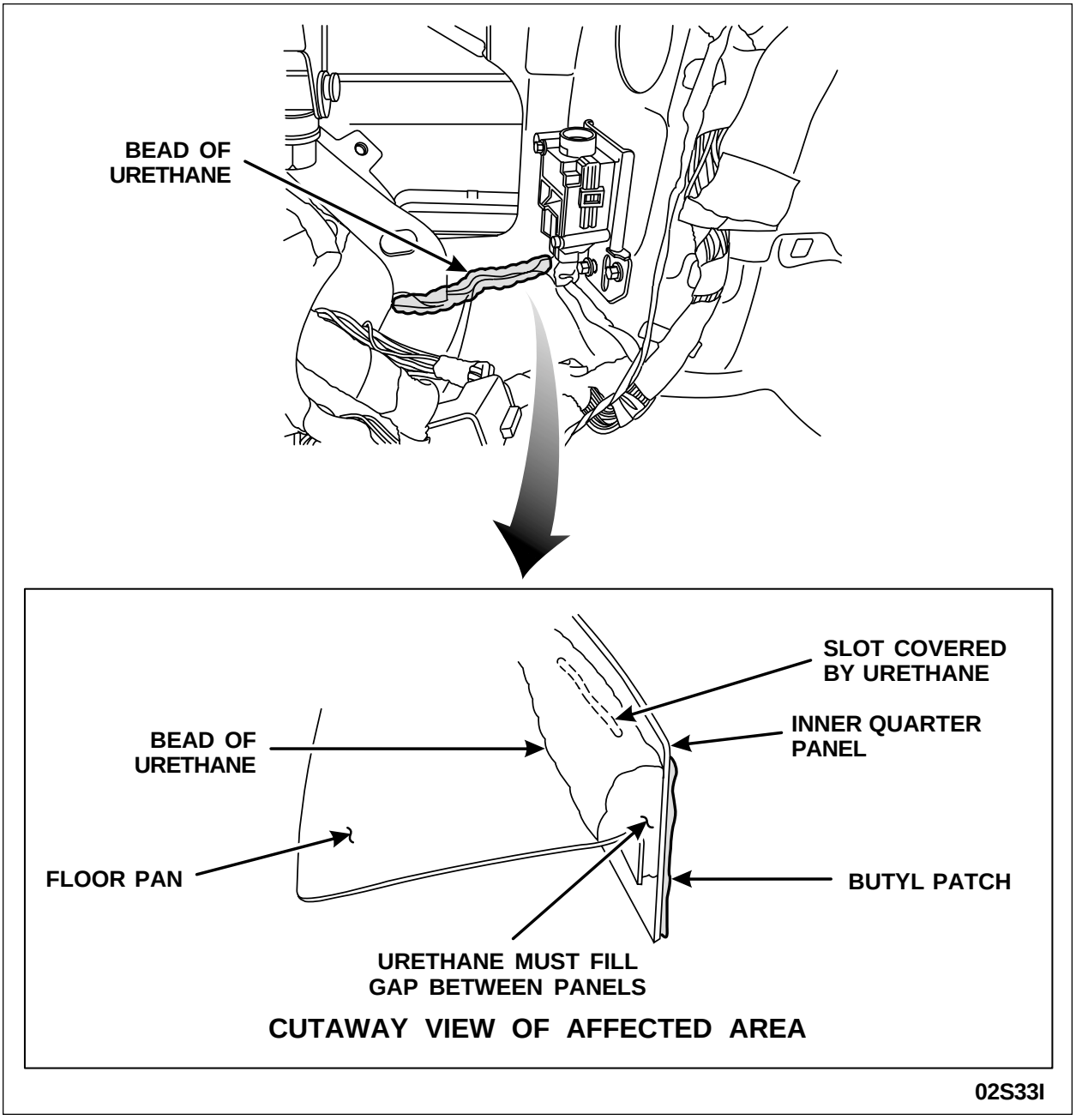


FIGURE 9

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4. Secure the butyl-wrapped bundle of splice connectors or trailer tow/RPA connectors to the vehicle wiring harness with vinyl tape. See Figure 10.

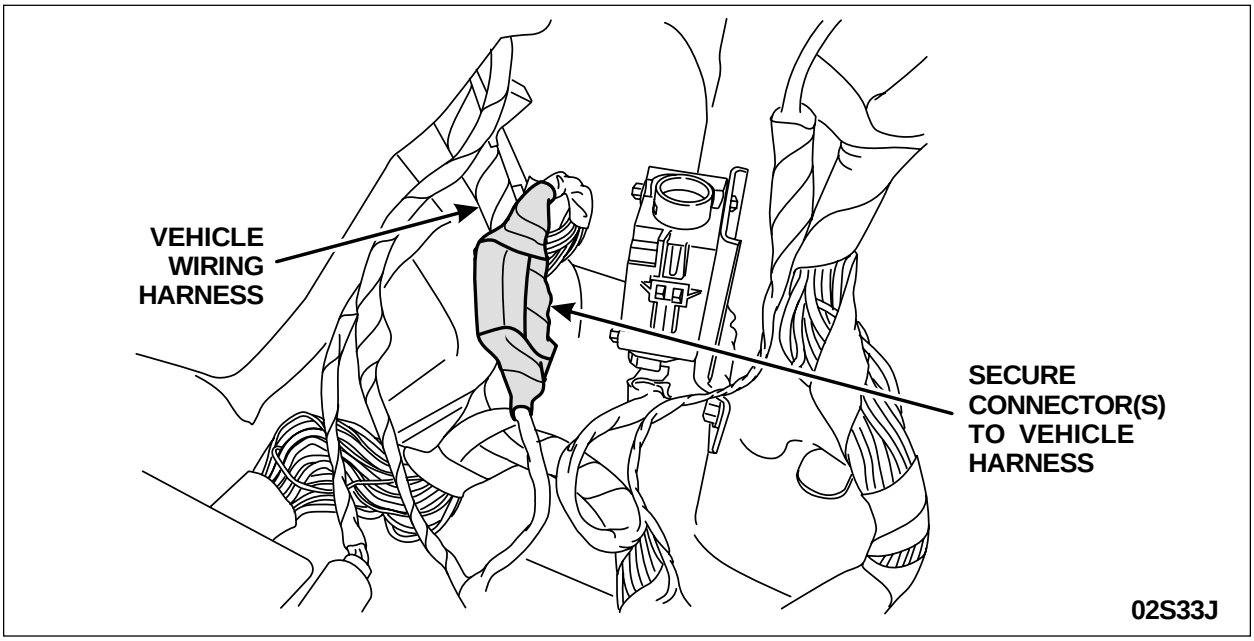


FIGURE 10

5. Install the insulation pack, if removed. Move the carpeting and watershield back to the original position.
6. Position the right-side lower D-pillar trim panel and connect the power point. Install the right-side lower D-pillar trim panel.
7. Install the right-side upper D-pillar trim panel and retainers.
8. **WARNING: BE SURE TO TIGHTEN THE SAFETY BELT ANCHOR BOLT TO THE PROPER TORQUE SPECIFICATION.**

   Using a safety belt bolt bit T77L-2100-A or equivalent, install the right-side D-pillar safety belt guide. Tighten the bolt to 40 Nm (30 lb-ft), then close the vinyl cover.
9. Install the liftgate scuff plate.
10. Install the third row bench seat (if equipped).
11. Install the right-side upper C-pillar trim panel.
12. **WARNING: BE SURE TO TIGHTEN THE SAFETY BELT ANCHOR NUT TO THE PROPER TORQUE SPECIFICATION.**

    Install the right-side C-pillar safety belt guide. Tighten the nut to 40 Nm (30 lb-ft), then close the vinyl cover.
13. Reposition the second-row seat.
14. Install the right-side sliding door scuff plate.
15. Connect the battery negative cable, then remove the memory saver.

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